



Auditing Fairness in AI-Based Resource Allocation

PS26 - AI, Power & Responsibility –
Governing Intel | Idea 3



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1 MOTIVATION



AI systems are increasingly used to allocate resources in critical domains.



Unfair allocations can disadvantage certain demographic groups, leading to inequity and mistrust



This study audits fairness, transparency and explainability in AI-based resource allocation systems.



Our goal is to build responsible AI that is not only accurate but also fair and accountable.

2 DATASET



German Credit Dataset



- 1,000 records
- 9 Predictive attributes + Demographics
- Binary target variable: Risk (1=Good Risk, 0 = Bad Risk)
- Protected Attributes:- Sex (Male, Female)



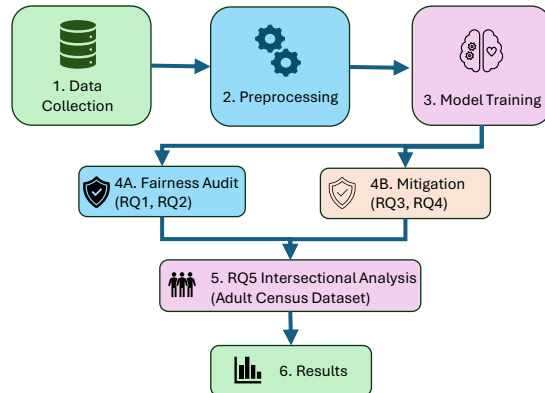
Adult Census Dataset

- 32,561 records
- Used for RQ5 (intersectional analysis)
- Protected Attributes: Sex, Race, Age

Key Features (German Credit)

Credit Amount, Duration, Purpose, Savings Account, Checking Account, Employment, Installment Rate, Personal Status, Age, Sex

3 METHODOLOGY WORKFLOW



4 RESEARCH QUESTIONS

RQ1

How do different ML models vary in fairness when predicting resource allocation outcomes across demographic groups?

RQ2

Which applicant characteristics contribute most to potentially biased resource allocation predictions?

RQ3

Do fairness mitigation techniques improve predictive fairness in AI-based resource allocation systems?

RQ4

What fairness-performance trade-off exists across different machine learning models?

RQ5

Do aggregate fairness metrics hide subgroup disparities revealed through intersectional analysis?

6 CONTRIBUTIONS



Comprehensive fairness audit of AI-based resource allocation systems.



Comparison of Logistic Regression, Random Forest, and XGBoost across performance and fairness metrics.



Explainability using SHAP to identify key drivers of biased predictions.



Evaluation of demographic bias across gender and age groups.



Actionable recommendations for building fair, transparent, and accountable AI systems.

7 CONCLUSION



Our study demonstrates that AI-based resource allocation models can exhibit demographic bias, and fairness varies across algorithms. While XGBoost provides the highest predictive performance, Random Forest offers the best fairness-performance balance.



Explainable AI techniques such as SHAP improve transparency by revealing the most influential factors in allocation decisions.



Integrating fairness auditing and mitigation strategies into AI systems is essential to ensure equitable and responsible decision-making in resource allocation.

5 KEY RESULTS (HIGHLIGHTS)

